

Summary Page

Proposal No: 5230**Title:** Utility of eye lens chemistry to inform the biology of Pacific sleeper sharks**Abstract:**

The Pacific sleeper shark (*Somniosus pacificus*) is a large-bodied shark species with broad geographic distribution in the North Pacific Ocean. The species is common as bycatch in commercial fisheries in Alaska and has been shown to be vulnerable to overfishing, yet little is known about even its basic biology and habitat use. The IUCN recently categorized populations in the North Pacific Ocean as decreasing and “Near Threatened”. There is a critical need for validated life history information to better inform stock assessments; however, traditional age determination methods have been unsuccessful. Eye lenses, which in vertebrates accrete layers continuously throughout life providing a chemical record of past conditions, have been successfully used in other hard-to-age species to provide independent records of longevity through bomb radiocarbon analysis. In this project we use radiocarbon analysis of sequential eye lens layers to inform age ranges and stable isotope and compound specific amino acid analysis to explore the ecology of the species. Preliminary results suggest that the species is long lived, with longevity likely >100 years.

Research Funded By: North Pacific Research Board**Priority:** Fishes & Fish Habitat**Region:** Bering Sea/Aleutian Islands**Presentation Type:** Oral**Student Award Eligibility:** This is not a student presentation**Presenting Statements:**

- Gave oral presentation last year: No
- Presented preliminary/interim findings last year: No
- Represents final report: Yes